



ARRAYS MCQs (Practice)

Date: / /

Q1. Which of the following is true about arrays in Java/C++?



REMEMBER:

- A) Indices start from 1. (X)
- B) Size must be defined at compile time (mostly). (?)
- C) They can store elements of different types. (X)
- D) Elements are stored in contiguous memory locations. (✓)

KEY PROP.

[✓ Correct Answer]

Q2. What is the result of 'arr[5]' for an array 'int arr[5] = {1, 2, 3};'?

REMEMBER:



- A) 3
- B) 0
- C) Compile Error (✓)
- D) Runtime Error.

arr[5] index out of bounds!
[Range: 0 to (size-1) = 0 to 4]

Q3. Which operation is $O(1)$ time complexity in an unsorted array?

V.IMP

- A) Access by Index (✓)
- B) Search for an Element
- C) Insertion at the beginning
- D) Deletion by Value

$O(1)$ - constant time!

[Requires traversal for B, C, D]

Arrays MCQs Part 2

14/10/2023

Arrays Practice Sheet - Set 2. Solve & revise! ☆

** Q1. Consider an array `int arr[5] = {10, 20, 30, 40, 50};`
What is the value of `arr[2] + *arr`?

A) 20

B) 40

☒ C) 60 ✓

D) 70

*Note: arr is base address.

*arr = arr[0] = 10

arr[2] is 3rd element = 30, 30 + 10 = 40

** Re-check base pointer
vs array indexing!!

** Q2. Which of the following statements about multi-dimensional arrays is incorrect in C++?

A) They are stored in Row-Major Order.

B) To access an element, you use `arr[row][col]`.

C) The last dimension must always be specified in function parameters.

☒ D) Elements within each row are not stored contiguously. ✓

* Incorrect stmt: Elements *are* stored contiguously in memory,
both within rows and across rows in Row-Major order.

** Q3. What happens if you try to access `arr[5]` for the array defined in Q1?

A) Gives garbage value.

B) Compilation Error.

☒ C) Runtime Error (e.g., Segmentation Fault). ✓

☆ D) Returns 0.

→ **CRITICAL!** Out of bounds!

Undefined Behavior (UB).

Likely Runtime Error or
Accessing random memory.

Compiler doesn't check
bounds.

** Remember: Index starts from 0 to n-1.

2D arrays: Row-Major vs Column-Major (rarely used).

Practice pointer arithmetic with arrays.

Arrays MCQs Part 3

1) What is the time complexity of accessing an element in an array using its index?

A) $O(\log n)$

B) $O(n)$

☒ C) $O(1)$ ✓

D) $O(n \log n)$

Remember: Index starts at 0!

2) Which of the following is correct about array initialization in C?

A) `int a[] = {1, 2, 3};`

B) `int a[3] = {1, 2, 3};`

C) `int a[3]; a = {1, 2, 3};`

☒ D) Both A and B

Size is static,
length is property

Correction

3) How do you find the size of an array 'arr' in Java?

☒ A) `arr.length`

B) `arr.length()`

C) `arr.size`

D) `arr.size()`

Size is static,
length is property